

Anirudh Iyengar Kaniyar Narayana Iyengar

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EDUCATION

Arizona State University

Master of Science in Robotics and Autonomous Systems - Artificial Intelligence

Tempe, AZ

December 2024

Dayananda Sagar University

Bachelor of Technology in Computer Science and Engineering

Bengaluru, India

May 2020

WORK EXPERIENCE

Data Systems Analyst Intern

January 2025 - Present

Local Grown Salads

- Establishing API endpoints using **FastAPI development** for IoT device state and sensor data management in a **PostgreSQL** database on **AWS** and deploying **Docker** containers for cross-system consistency and scalability.

Data Analyst

June 2024 - December 2024

Axyo (Synapse Labs Inc.)

- Collected, cleaned, and analyzed structured and unstructured financial data to streamline reconciliation processes; automated ETL pipelines to **reduce manual efforts by 70%**.
- Designed **visual dashboards and KPIs using Amazon QuickSight** for key internal stakeholders, driving strategic decision-making and improving reporting efficiency.
- Collaborated cross-functionally to support internal data needs for audits and compliance, **leading to a 10% revenue impact** through enhanced transparency and data accessibility.
- Developed **data processing pipelines and archival systems using AWS and containerized environments** (Docker, Kubernetes), ensuring secure and scalable data operations.

Data & Research Assistant

July 2023 - June 2024

ASU College of Health Solutions, JLiAng Lab

- Conducted statistical and machine learning analyses on healthcare data using **Deep Neural Networks, CNNs, and Transformers** in PyTorch, collaborating with Valleywise Health to improve accuracy and efficiency.
- Fine-tuned **NLP models (GPT-4, CLIP, RAM++)** on electronic healthcare reports using **Qdrant - Vector Database** for query classification, boosting performance by **5%**, and presented model benchmarks with **Power BI** dashboards.

Data Analyst

January 2021 - December 2022

HIB

- Initiated and led an OCR-based pipeline using **PyTesseract** to digitize 150,000 handwritten bills, minimized manual data entry time by **40%**; utilized **Python, SQL, and Pandas** for data preprocessing, entity resolution, and key-value extraction.
- Applied **XGBoost and Random Forest** regression models to identify daily product trends, refining pricing strategies and generating weekly reports, achieved **15% revenue growth** and a **10% increase in customer satisfaction**.
- Programmed a customer segmentation model using **K-Means clustering**, identifying high-value customer groups and amplifying marketing campaigns, leading to a **7% increase in repeat purchases** and improved campaign ROI.
- Designed **Tableau** interactive dashboards to track and visualize sales trends, reducing **weekly analysis time by 10%**.

PROJECTS

AI Hotel Reservation System | *LightGBM, MLflow, Jenkins, GCP, Data analysis, Machine Learning.*

Present

- Integrated a hotel reservation prediction system with **5% improved accuracy** using **LightGBM, MLflow** for tracking, **Jenkins** for CI/CD automation, and deployed on **Google Cloud Run** for efficient and scalable inference.

Integration of RAG with Open Source LLM and LangChain | *RAG, NLP, Quadrant, LangChain*

Present

- Enhanced source accuracy by 2% using **Qdrant, BGE-large-en-v1** embeddings, and **LangChain with LLM (BERT)** in an optimized pipeline for seamless **text generation in Q&A and research summaries**.

Mapping Accident Trends and Patterns in Maryland | *D3.js, Javascript, HTML5, CSS3/SCSS*

December 2024

- Developed visualizations like **Car-in-a-Clock, Mosaic Chart, Tree Map, Geo-Spatial Mapping, and Pie Matrix** for The PacificVis Storytelling Contest, with enhanced interactivity via scroll effects, hover actions, and tooltips.

TECHNICAL SKILLS

Data Analysis & Visualization: Excel, PowerPoint, Tableau, AWS QuickSight, Power BI, Matplotlib, Seaborn, Pandas, NumPy.

Languages: Python, C, C++, SQL, HTML, MATLAB, Bash, R, CSS, JavaScript, D3.js, TypeScript.

Database Management: PostgreSQL, MySQL, MongoDB, DynamoDB

Tools & Platforms: Microsoft Office Suite, Git, Jupyter, FastAPI, Docker, Google Cloud (BigQuery, Cloud Storage), AWS (S3, SageMaker), Databricks, CI/CD Pipelines, MLflow, Jenkins, Airflow, Visual Studio Code.